

A deterministic boundary counterexample to conjecture 00000003479

earthking11

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Statement and interpretation

The conjecture defines a harmonious coloring to be a proper coloring in which sums of endpoint color labels distinguish all unordered edges. It claims that the harmonious chromatic number $h(G(n, p))$ concentrates at $(1 + o(1))\sqrt{np}$, uniformly for $p \geq \log n/n$. We use precisely the sum definition, not the different standard definition based on unordered color pairs. Colors in a k -coloring have natural labels $0, \dots, k - 1$. The argument uses only properness, so any additional sum restriction cannot rescue the claimed upper concentration.

Lemma 1. *Every harmonious k -coloring of K_n has $k \geq n$.*

Proof. Every two distinct vertices of K_n are adjacent. Properness therefore makes the color map injective. An injection from n vertices into k colors implies $n \leq k$. $\square$

Theorem 1. *For every integer $n > 4$, the probability that $G(n, 1)$ admits a harmonious k -coloring with $k \leq 2\sqrt{n}$ is zero. Consequently, the stated constant-one concentration law is false.*

Proof. At $p = 1$, every edge is present with probability one and $G(n, 1) = K_n$ almost surely. This parameter is admissible for every $n \geq 1$, because $\log n \leq n - 1 \leq n$. By the lemma, a coloring in the asserted event would give $n \leq k \leq 2\sqrt{n}$. Squaring implies $n^2 \leq 4n$, which contradicts $n > 4$. Thus the upper event has probability exactly zero for all $n > 4$.

In particular, if h is the minimum number of colors, the relative concentration event

$$\left| \frac{h(G(n, 1))}{\sqrt{n}} - 1 \right| < 1$$

is contained in that impossible upper event. Its probability is eventually zero and tends to zero, not one. The conjectured relative concentration would require convergence to one for every positive tolerance, including this tolerance. Failure at the admissible deterministic boundary refutes the claimed uniform law. No finite-size numerical anchor is used. $\square$

Formal verification and scope

The Lean project uses Mathlib's actual `SimpleGraph`, graph `Coloring`, and probability mass functions. It defines harmonious colorings with the endpoint-sum condition. For every finite graph it constructs a coloring with labels $(2n^2 + 1)i + i^2$: a label-pair sum determines the index sum and square sum, hence its unordered endpoint pair. Thus a minimum exists. Lean defines the minimum by natural-number well-ordering, proves its attainment and minimality, and identifies its relative-error event with the formally defined concentration event. The definition `GnpOneLaw` specifies that each distinct edge has probability one; Lean proves every graph in its support is K_n . The explicit point mass at K_n satisfies this law. Thus the proof does not assume a counting bound or a random-graph tail estimate. The capstone `conjecture_boundary_is_false` proves the negation of constant-one relative concentration at $p = 1$. All statements compile on Lean 4.33.1 with pinned Mathlib, without unfinished proofs, native evaluation, or additional axioms. The only reported foundational axioms are propositional extensionality, classical choice, and quotient soundness. We do not claim a corrected concentration law for $0 < p < 1$, nor need the conjecture's additional low-probability clause.